Virtual Radio Direction Finding

USER MANUAL



Both SSB tracking mode for Egg hunt & FM tracking mode for Fox hunt.

Options for Multiple Hound's & Multiple Fox's available.

Use your radio to communicate with other players.

Use your Yagi directional antenna, Map & Compass to find the Egg/Fox as quick as you can.

Welcome

Virtual Radio Direction Finding is a multiplayer first-person game about hunting hidden radio transmitters, a hobby known as **radio direction finding** or **foxhunting**. You carry a handheld radio for talking to your friends and a three-element **Yagi** directional antenna for tracking down the signal. Two kinds of game are played on a big map of hills, forest and villages: the **Radio Egg Hunt** (find a hidden beacon, fastest wins) and the **Radio Fox Hunt** (hounds track down hiding foxes).

Contents

Getting started 2

1 Radio Egg Hunt 4

2 Radio Fox Hunt 6

3 The Yagi Directional Antenna 9

4 The Radio Voice Service 11

5 Administrator Features 13

Getting started

- **Start the game** with **Game.exe**. It connects to the game server on its own. To use a different server, start it with the address, for example `Game.exe 192.168.1.5` or `Game.exe myserver.example:7777`.
- **Log in** with your player name (3 to 15 letters, digits or _) and password (4 to 31 characters). If the name does not exist you are asked whether to **create the account** or try again.
- **The lobby** lists everyone online. Click **Radio EGG hunt** or **Radio FOX hunt** and everyone in the lobby joins that game. Players who log in while a game is running wait in the lobby for the next one. **Key binds** lets you choose your own keys and mouse buttons, and **Scoreboard** shows the results.

Default controls (all can be changed in Key binds)

Action	Default key	What it does
Walk	W A S D	Forward, left, backward, right. Moving diagonally is 50% slower.
Look around	Mouse	The cursor is hidden and captured while you are in the world.
Run	Shift (hold)	Up to 2.5 times walking speed. Uses endurance (see below).
Raise the Yagi	Space (hold)	Lets you hear the beacon or the fox. Lower it to stop listening.
Map	M (hold)	Shows the whole map with you as a yellow arrow, and your X / Y position above it.
Compass	C (hold)	A hand-held compass: the red end of the needle always points north.
Hide mode	H	Foxes only: lock yourself in place when you are hidden.
Talk on the radio	Left mouse button (hold)	Push-to-talk. Release to listen.
Quit	Esc	Opens "Quit game - Are you sure?" (answer Y or N). Fixed, cannot be changed.

Endurance. A bar at the top right shows your endurance while you run. Running drains it by **2% per second**; when you are not running it recovers by **3% per second**. Your running speed depends on how much endurance is left, and at 0% you cannot run until you let go of the run key. Tree trunks and the walls of houses are solid

(houses have an open doorway), but grass and shrubs can be walked straight through.

Scoreboards. The Scoreboard screen is split in two. The **top half** shows the game in progress, or the last game, and updates live as players finish. The **bottom half** has tabs holding the **permanent** best times: *Egg Hunt*, *Single Fox Hunt* and one tab each for 2 to 8 foxes. If the last game made a new high score, that tab flashes red, yellow, green and blue until you click it.

1 Radio Egg Hunt

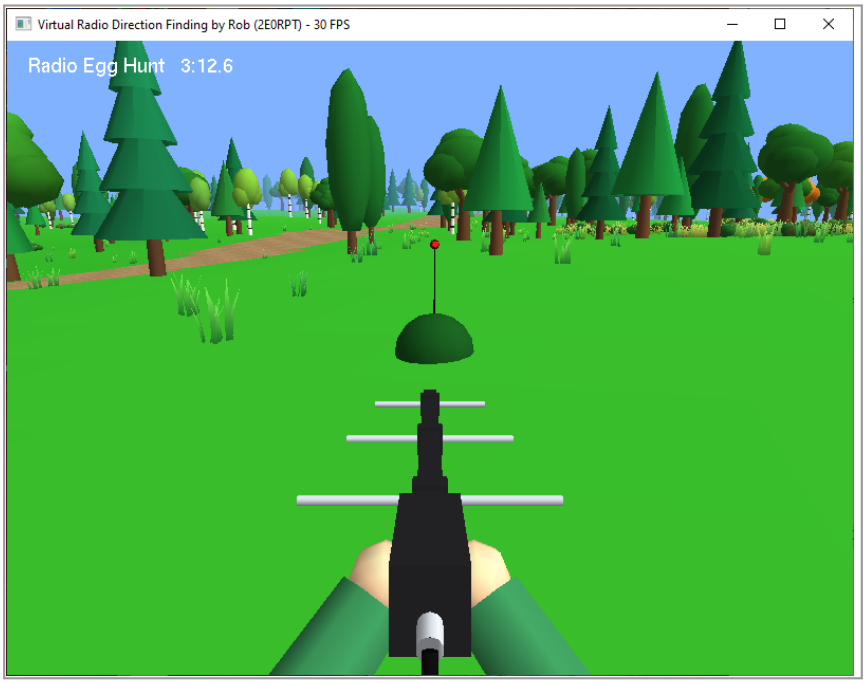
Find the hidden beacon as fast as you can

The aim

A small radio beacon, the **egg**, is hidden somewhere on the map. It is a **small dark green dome with a thin black antenna and a red light on top**, and it is never shown on the map. Everyone in the game is dropped at a random place at least 300 units from it and has to track it down by ear. Walk into the beacon and your round ends. Your time is recorded and you go to the scoreboard to wait for the others.

How a round works

- **Start.** Click **Radio EGG hunt** in the lobby. A **3 second countdown** gives everyone a moment to look around, then the clock starts.
- **Hunt.** Hold the Yagi up (**Space**), sweep it round, walk towards the loudest beep, and use the map and compass to keep track of where you are.
- **Find it.** When you touch the beacon the round ends for you. You are placed **1st, 2nd, 3rd...** in the order you arrive, with your time.
- **Wait.** The scoreboard updates live as the others finish. When every player has found the beacon or left the game, a **Back to lobby** button appears.



*Tracking the egg: the Yagi is raised and pointing at the beacon. Close up, the beacon becomes visible and its **red light flashes with every beep**. Far away it is hidden: the game only shows it when you are within about 200 units.*

How Egg Hunt sounds: SSB tracking

The Egg Hunt is modelled on **SSB (single-sideband) tracking**, the way direction finders listen for a weak carrier. While the Yagi is raised you hear a **constant background noise** of static (a mix of hiss and rumble). Every second a short **1000 Hz beep** pokes through it. The noise never changes, but the **beep gets louder** as you:

- **point the Yagi directly at the beacon** (see section 3),
- **get closer** to it, and
- **stand on higher ground**.

The beep volume is worked out from three things multiplied together: **direction volume x distance volume x elevation volume**.

Part	How it works
Direction	Yagi pattern: 100% pointing straight at the beacon, 20% when it is 90 degrees off to the side, silent pointing directly away.
Distance	100% right next to the beacon, falling in a straight line to 0% at the map corner farthest away from it.
Elevation	10% on the lowest ground up to 100% on the highest hilltop (but never lower than the distance volume, so a close beacon is always clear).

Example

Worked example. You stand 600 units from the beacon. Its farthest corner is 1800 units away and the ground under you is 6.0 units high. You aim 30 degrees off the beacon. Direction = 85.1%, distance = 1 - 600/1800 = 66.7%, elevation = 78.2%. The beep you hear is 0.851 x 0.667 x 0.782 = **44.4%** of full volume.

Hunting tips

- **Triangulate.** Face the loudest direction, hold **M** and note where the arrow points. Walk well to one side, repeat, and the two lines cross at the beacon.
- **Sweep in steps.** The beacon beeps once a second and the volume is fixed at the start of each beep. Turn 15 to 20 degrees, stop, listen to one beep, then turn again.
- **Find the middle of the loud zone.** The volume changes slowly near the peak. Find the two angles either side where the beep is clearly half as loud (about 61 degrees each way) and aim halfway between them.
- **Climb before you listen.** On the lowest ground the beacon is only a tenth as loud as on the highest hilltop.
- **Quiet does not mean the wrong direction.** Distance and low ground make everything quieter. Compare beeps from the same spot as you turn.
- **Save your running** for open ground and walk while you sweep, so your endurance refills while you listen.
- **Where it is not.** The beacon is never on a path, never close to a house and never near a tree.

2 Radio Fox Hunt

Hounds track down the hiding foxes

The aim

In a Fox Hunt there is no beacon. Instead some players are **foxes**: they run off, find a hiding place and then transmit on a timetable. The other players are **hounds** and use their Yagi to find them. The round lasts until **every hound has found every fox**.

Setting up: the Fox Hunt lobby

Click **Radio FOX hunt** in the main lobby (it needs **at least two players** in the lobby). Everyone in the lobby moves to the Fox Hunt lobby, where each player has three buttons beside their name: **Hound**, **Fox** and **Ready**. A button is **light green with black text when it is switched on** and **red with white text when it is off**. Everyone starts as a Hound.



Choose your role, then press **Ready**. Changing your role clears your Ready.

Below the player list are two settings with **- / +** buttons:

- **How long should each fox TX for**: 10 to 60 seconds, in steps of 5.
- **How long does the fox(s) wait to TX again**: 15 to 900 seconds, in steps of 15.

The green **Everyone ready? - Let's play** button works only when there is **at least 1 hound and 1 fox** and **every player has pressed Ready**.

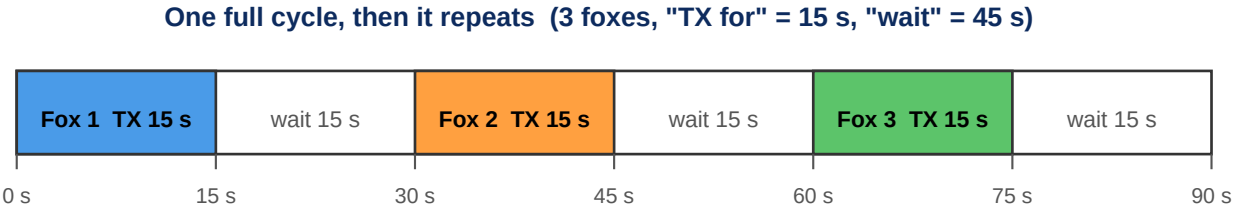
*The Fox Hunt lobby set up for **three foxes** (Bob, Bill and Jim) and one hound (Rob). Each fox will transmit for 15 seconds, with a 45 second wait shared between them.*

The hiding phase

- The foxes are dropped at random places on the map and have **3 minutes** to find somewhere to hide. During this time foxes have **infinite endurance**.
- A fox that is happy with its spot can press **H** to enter **hide mode** early, or simply wait for the 3 minutes to run out. In hide mode a fox stays where it is.
- The hounds wait on a "The foxes are hiding..." screen with a countdown.
- When the 3 minutes are up, or **all** the foxes have pressed H, the hunt starts. The foxes turn **mostly transparent** so they blend into their surroundings, and the hounds appear together at the **centre of the map**, standing side by side.

The fox transmissions: timers and the pie chart clock

A fox does **not** choose when to transmit. The server runs the timetable from the two settings you chose in the lobby. **The first transmission starts 15 seconds after the hunt begins.** When there is more than one fox they take turns, and the **wait time is divided evenly between them**: Fox 1 transmits, then there is a gap of (wait / number of foxes), then Fox 2, and so on.



The 45 s wait is divided evenly between the 3 foxes: one 15 s gap after each fox.

If you are a fox, a message counts down to your next turn ("**Next transmission in 1:25**") and switches to a flashing **ON-AIR** with the seconds you have left while you are transmitting. Speak clearly while watching the ON-AIR time remaining, Talk to or taunt the hounds, But never be rude, If you don't speak or your microphone is not working correctly a warble tone will be transmitted instead.

If you are a hound, a **pie chart clock** sits at the top right of the screen. It shows one complete cycle starting at 12 o'clock and going clockwise:

- each coloured slice is one fox's transmission (**Fox 1** blue, **Fox 2** orange, **Fox 3** green...), as wide as the "TX for" time,
- the **white** parts are the waits, and
- a **red dot** travels clockwise round the rim. When it is inside a coloured slice, that fox is on air; when it is in the white, nobody is.

Under the clock a caption says who is on air ("Fox 1 on air - 20 s") or how long until the next transmission. The message **"A fox is on air - raise your Yagi (SPACE)"** flashes while a fox transmits. The clock re-aligns with the server at the start of every transmission, so it never drifts.



A hound's view during the hunt. Rob has his Yagi raised. Fox 1 (the faint figure by the tree) is on air, and the pie clock shows the red dot inside the blue Fox 1 slice. Foxes are almost invisible and only show up when you are close.

How Fox Hunt sounds: FM tracking

The Fox Hunt is modelled on **FM tracking**, which behaves very differently from the Egg Hunt's SSB. Instead of a beep getting louder, the **noise goes away**:

- When the Yagi is **pointing away** from the fox or you are **far away**, you hear mostly **hissy static**.
- As you **get closer and point the Yagi at the fox**, the **noise drops and the fox sounds clearer** and cleaner.
- When a fox **starts** a transmission you hear a short **squelch burst** of noise. When it **ends** you hear that fox's own **5-note roger beep** followed by another squelch burst.
- With several foxes you can **tell them apart** by their roger beeps, and the pie clock tells you who is due next.
- You only hear the fox **while the Yagi is raised**. Lower it and the world goes quiet again. The fox is **never** heard on your voice radio, so the radio cannot be used to find a fox (section 4).

What does a fox sound like? Normally a steady, slightly **warbling tone**. But the fox's microphone is open while it is on air: if the fox **talks**, the hounds hear the **fox's own voice instead of the warble**. If nobody has spoken for 3 seconds the warble comes back, and it switches to voice again the moment the fox talks. Only foxes that are on air are heard, and the voice goes through the Yagi just like the tone: clean when you aim at it, hissy when you do not.

Fox microphone controls

Every microphone is different, so a fox in hide mode gets a small panel at the top right of the screen with a **live microphone level meter** and a **Voice level slider** of the same size. The white line across the meter is your voice threshold (**default 20%**). When the meter goes above the line, the game counts you as talking. Drag the slider so that your normal speaking voice goes above the line but background noise stays below it.

> **Fox in hide mode**
> In hide mode the mouse pointer is released so you can use the slider. The mouse can no longer turn the camera, so the **walking keys (W A S D) pan the camera** instead. Your microphone is only sent to the hounds while you are on air.

Finding a fox

Walk right up to a fox to find it. A message tells you "You found fox 1!" and the HUD counts **Foxes found: 1 / 3**. Once you have found them all you can stay on the map and help (or hinder) the others. The round ends when **every hound has found every fox**, or when no hounds or no foxes are left. Hounds are ranked by the time it took them to find all the foxes. Each size of Fox Hunt (1 to 8 foxes) has its own permanent scoreboard tab.

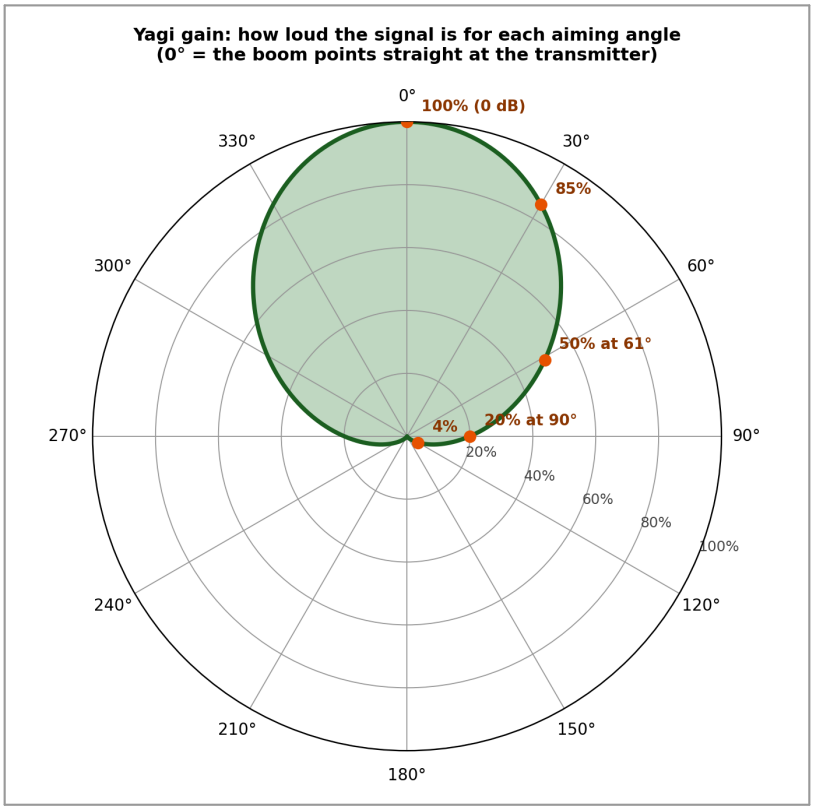
3 The Yagi Directional Antenna

Your most important tool

What it is

A **Yagi** is a directional antenna with a long boom and several cross elements. It receives best along the direction the boom points (the **main lobe**) and much less from the sides and rear. In the game you hold a **three-element Yagi**. Hold the **Space** key to raise it and listen, and let go to lower it. While it is raised you can still walk, and the voice radio and the ambient background sound change as follows: the handheld radio display is hidden while the Yagi is up (you can still hear your friends), and the background sound loop pauses until you lower it again.

Only the **compass direction** you are facing matters: looking up or down does not change the signal.



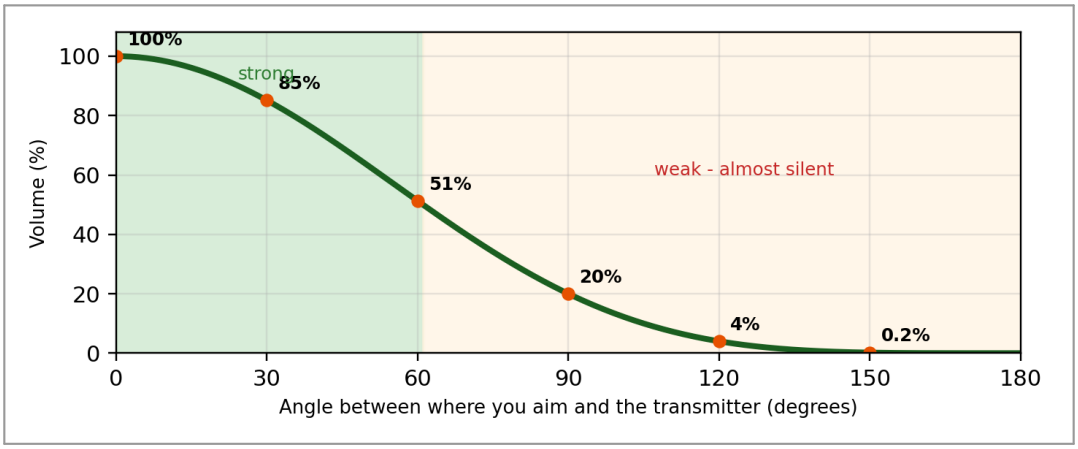
Yagi gain pattern (polar plot). The further the green curve is from the centre, the louder the signal. Straight up is the direction the Yagi points at the transmitter.

Gain in every direction

The volume you hear for the direction part of the signal is `((1 + cos(angle)) / 2) ^ 2.32`, where *angle* is how far the boom is from the transmitter. It does not matter whether you are off to the left or to the right: the same angle gives the same volume.

Angle off target	Volume	Level (dB)	What you hear
0°	100%	0.0	Full volume
15°	96%	-0.3	Almost full
30°	85%	-1.4	Strong
45°	69%	-3.2	Clearly quieter
60°	51%	-5.8	About half
90°	20%	-14.0	Weak

Angle off target	Volume	Level (dB)	What you hear
120°	4%	-28.0	Very faint
135°	1.2%	-38.7	Barely there
150°	0.2%	-54.5	Almost nothing
180°	0%	minus infinity	Silent (null)



The same gain as a curve: the signal falls away smoothly as you turn off target, and reaches silence pointing directly away.

Using the Yagi well

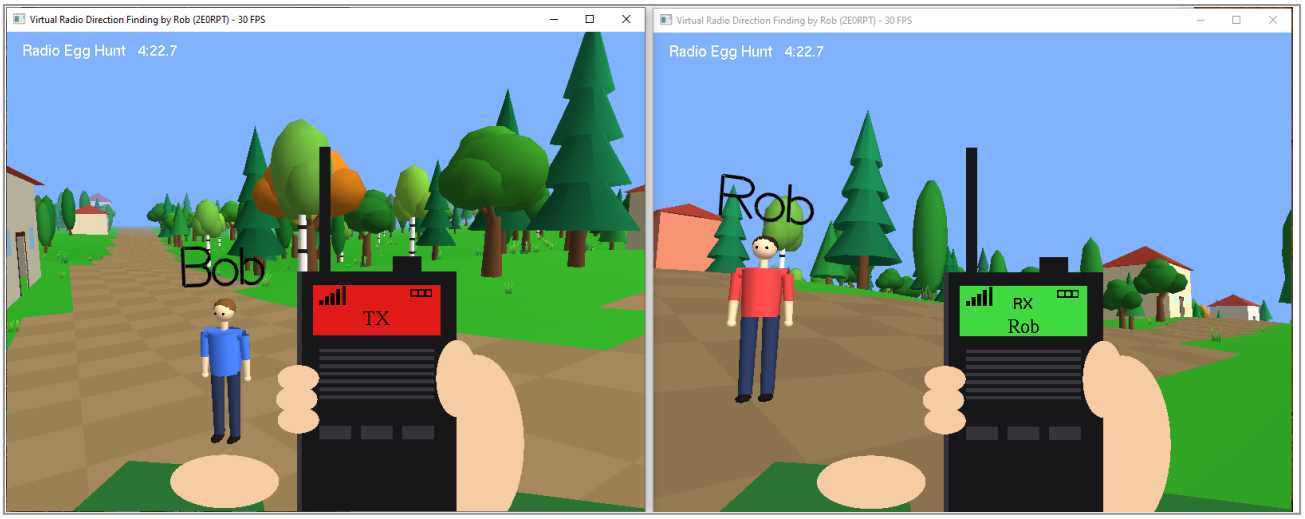
- **The beam is about 122 degrees wide.** The signal is half as loud about **61 degrees** either side of the centre line, so the usable main lobe is roughly 122 degrees wide.
- **The peak is flat, the sides are steep.** At 15 degrees off, the volume is only 4% lower, so it is hard to judge the exact top. Instead find the two angles either side where the signal is clearly **half** as strong and aim **halfway between them**.
- **The game's Yagi is idealised.** It has a perfect silent point directly behind it and no side lobes. A real three-element Yagi picks up some signal from the back and sides, which is why real foxhunters often use an attenuator.
- **Combine it with the map and compass.** Take a bearing with the Yagi, check it against the map (**M**) or compass (**C**), move, and take another. The point where your lines cross is the transmitter.
- **In the Egg Hunt** the Yagi changes how loud the beep is; **in the Fox Hunt** it changes how noisy the signal is. In both, the strongest or clearest direction is the way to go.

4 The Radio Voice Service

Talk to your friends, or keep them out of earshot

Your handheld radio

Besides the Yagi, every player has a second radio for **talking to other players**. It works like a real push-to-talk handheld: **hold the left mouse button to talk, and release it to listen** (you can change the button in Key binds). It is separate from the Yagi and the beacon: it is not directional and **cannot be used to find an egg or a fox**.



Two players talking.

Left: Rob holds the talk button and his radio shows a red TX screen.

Right: Bob hears him and his radio shows a green... RX screen with the name of the player talking.

What you see on the radio

- **Transmitting:** a **red screen with "TX"**.
- **Receiving:** a **green screen with the name of the player talking**. The signal bars on the screen show how clear the signal is.
- The radio only appears while someone is talking. It is **hidden while your Yagi, compass or map is raised**, but you can still hear everything.
- If an administrator has switched your voice off, the radio shows **VOICE OFF** when you try to talk.

How the radio sounds

- **One person talks at a time.** While someone is transmitting, others have to wait. No one may hold the channel for more than **30 seconds**; after that the radio cuts them off and shows **TIME OUT**.
- **Roger beep.** When you let go of the button your radio ends the "over" with a **5-note roger beep** made of tones from a pentatonic scale, followed by a short **squelch burst**. Every player gets **their own, unique roger beep**, chosen at random when your account was created and kept for good, so you can tell who has just finished speaking.
- **Static grows with distance.** In the Egg Hunt, the further apart two players are, the more static is mixed into their voices: none when standing together, and full static at opposite corners of the map.
- **No distance clues in the Fox Hunt.** The radio is completely **omni-directional** there and carries no distance static, so it can never give a fox away.

Talking in the lobby, on the scoreboard and in the fox lobby

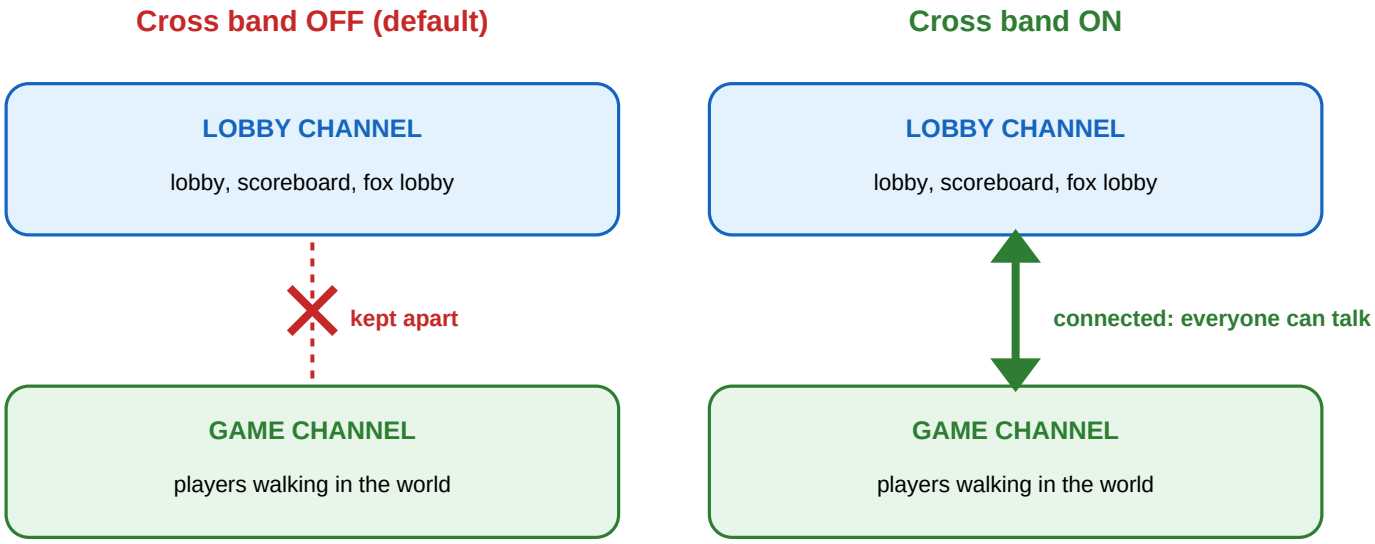
The radio is not only for use in a game. You can **talk and listen in the lobby, on the scoreboard and in the Fox Hunt lobby** as well. Menu players talk on their own **lobby channel**, and players who are walking around in

a game talk on the **game channel**. The two are kept apart so that chat in the lobby never disturbs players in the middle of a round, and finished players cannot give away the beacon or the foxes.

In the menus a small badge in the corner shows a red **TX - talking** or a green **RX: name**, or reminds you which key to hold to talk. If the push-to-talk button is also a menu click, holding it over empty space talks, while clicking a button works as normal.

Cross band

The **Cross band** check box in the lobby joins the two channels together. When it is **ticked**, players in the lobby, on the scoreboard and in the fox lobby can **talk with the players who are still playing the game**, and the other way round. When it is **not ticked** (the default), the two channels stay separate and lobby chat does not affect the game.



Good to know

Any player can tick or untick Cross band, unless an administrator has **locked** it (it then shows "Cross band (locked)"). A person in a menu has no position on the map, so their voice always arrives without distance static.

5 Administrator Features

Running a fair and friendly server

Becoming an administrator

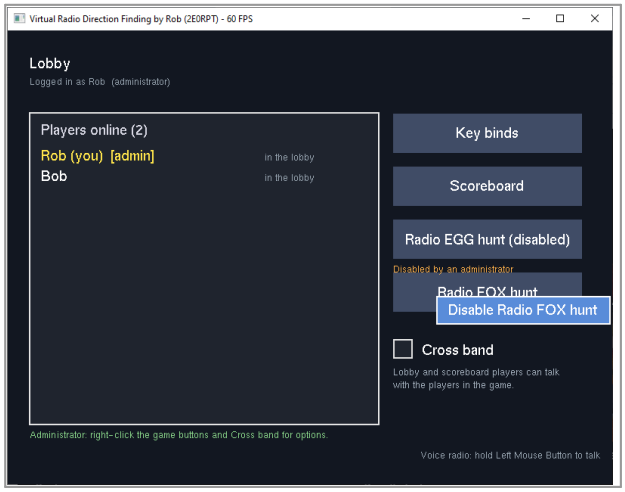
Administrator status can **only** be given or taken away at the **server console** (the window where Server.exe is running). It cannot be changed from the game, so players cannot give it to themselves. Type these commands in the server window:

Console command	What it does
`admin add NAME`	Makes that player an administrator (immediately, even if they are logged in).
`admin remove NAME`	Takes their administrator rights away.
`admin list`	Lists the administrators and shows who is logged in.
`kick NAME`	Disconnects that player from the server. They **can log in again**.
`ban NAME`	Bans the account. They **cannot log in** until they are unbanned.
`unban NAME`	Lifts a ban. **Un-banning is only possible at the server console.**
`banned`	Lists the banned accounts.
`help`	Shows the list of commands.

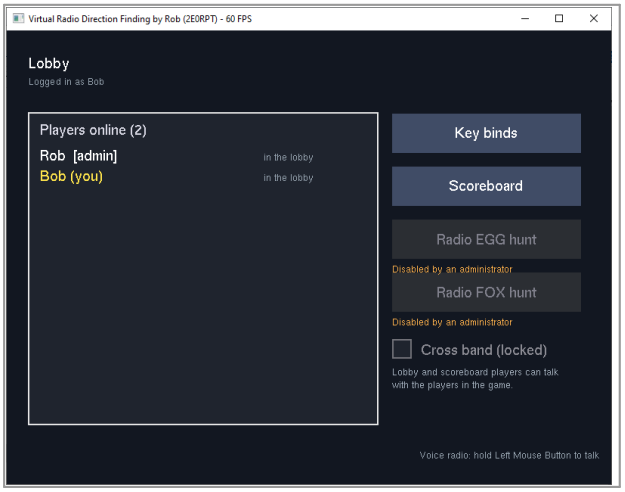
An administrator is marked **[admin]** next to their name in the player lists, and a short hint in the lobby reminds them to right-click for options. Everything below is done by **right-clicking**.

Switching a game off: the Egg and Fox buttons

In the lobby, an administrator can **right-click "Radio EGG hunt" or "Radio FOX hunt"** to get a pop-up that **disables or enables** that button. When a game is disabled, other players cannot click it and see **"Disabled by an administrator"**, but the other game is still free to use. Administrators can still start a game they have disabled.



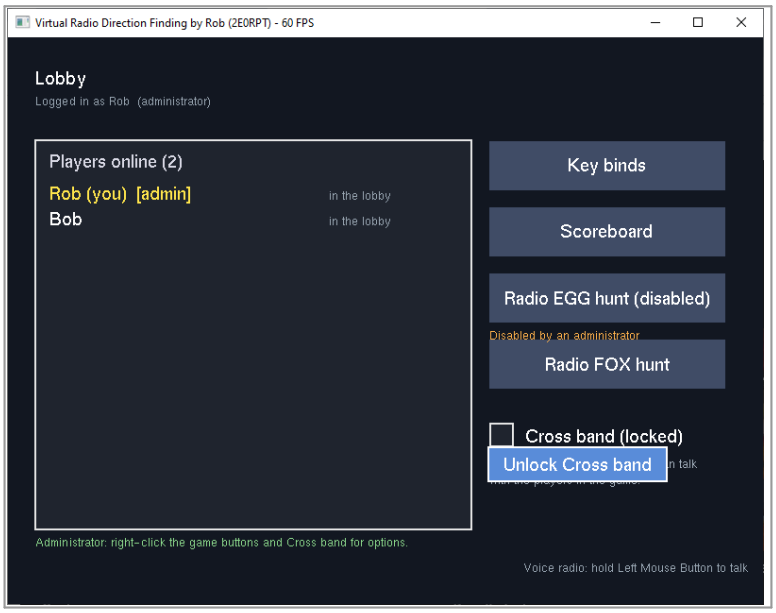
The administrator has disabled Radio EGG hunt (it shows "(disabled)") and is right-clicking Radio FOX hunt for the same option.



What an ordinary player sees: both game buttons greyed out, "Disabled by an administrator", and Cross band locked.

Locking Cross band

Right-click the **Cross band** check box and choose **Lock** (or **Unlock**) so that other players cannot turn it on or off. The label changes to "Cross band (locked)". Administrators can still change it.



Right-clicking the check box offers "Unlock Cross band" because it is currently locked.

In the Fox Hunt lobby

- **Set anyone to Hound or Fox.** An administrator can click the Hound and Fox buttons on **every** player's row, not just their own. That player's Ready is cleared so they can confirm again.
- **Lock the timers.** Right-click one of the **+ or - buttons** and choose **Lock the + / - buttons for other players**. The two settings can then only be changed by administrators, and everyone sees "(locked)".



The Fox Hunt lobby from an administrator's point of view, with the pop-up for locking the + / - buttons.

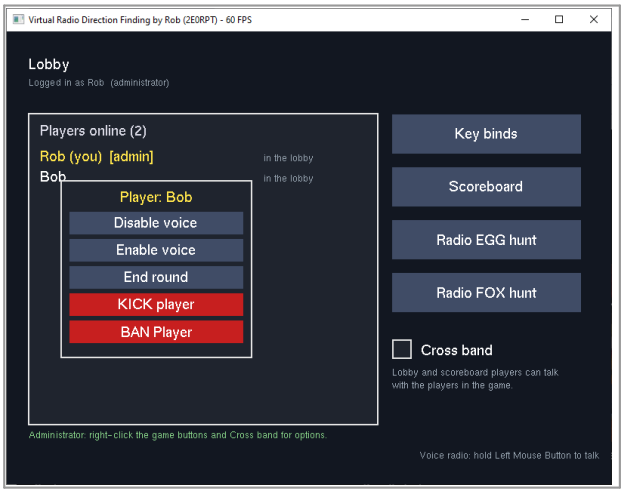
Automatic release

Lock-outs do not last forever. If **no administrator has been logged on for 2 minutes**, every lock-out (disabled game buttons, Cross band lock and the + / - lock) is lifted automatically and all players get full control again.

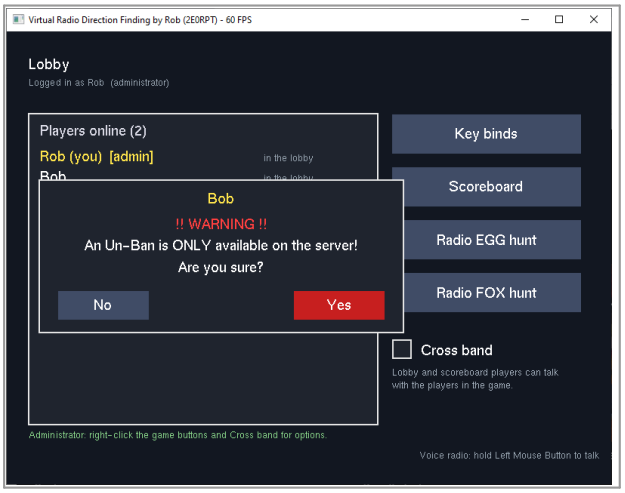
The player menu

Administrators can **right-click another player** in the **Lobby**, the **Fox Hunt lobby** or on the **Scoreboard** to open a menu with five options:

Option	What it does
Disable voice	That player can no longer talk on their radio (their radio shows VOICE OFF). Lasts until they disconnect.
Enable voice	Lets them talk again.
End round	Pulls the player out of the current game. They go to the scoreboard marked **"did not finish"**.
KICK player	Asks "Are you sure?" with **No** and **Yes**. Yes disconnects them from the server. They can log in again.
BAN Player	Asks for confirmation with a warning. Yes bans the account and disconnects the player.



Right-click a player's name to open the player menu. KICK and BAN are shown in red.



Choosing BAN Player shows a warning: an un-ban is ONLY available on the server.

Please note

An administrator cannot be kicked, banned or silenced from this menu, and you cannot use it on yourself. A banned player sees "This account is banned from the server" when they try to log in; bans are saved on the server, and only `unban NAME` at the server console removes one. A kicked player sees "You were kicked off the server by an administrator." and can log straight back in.